

C. Checkposts

time limit per test: 2 seconds

memory limit per test: 256 megabytes

Your city has n junctions. There are m **one-way** roads between the junctions. As a mayor of the city, you have to ensure the security of all the junctions.

To ensure the security, you have to build some police checkposts. Checkposts can only be built in a junction. A checkpost at junction i can protect junction j if either $i = j$ or the police patrol car can go to j from i and then come back to i .

Building checkposts costs some money. As some areas of the city are more expensive than others, building checkpost at some junctions might cost more money than other junctions.

You have to determine the minimum possible money needed to ensure the security of all the junctions. Also you have to find the number of ways to ensure the security in minimum price and **in addition in minimum number of checkposts**. Two ways are different if any of the junctions contains a checkpost in one of them and do not contain in the other.

Input

In the first line, you will be given an integer n , number of junctions ($1 \leq n \leq 10^5$). In the next line, n space-separated integers will be given. The i^{th} integer is the cost of building checkpost at the i^{th} junction (costs will be non-negative and will not exceed 10^9).

The next line will contain an integer m ($0 \leq m \leq 3 \cdot 10^5$). And each of the next m lines contains two integers u_i and v_i ($1 \leq u_i, v_i \leq n$; $u_i \neq v_i$). A pair u_i, v_i means, that there is a one-way road which goes from u_i to v_i . There will not be more than one road between two nodes in the same direction.

Output

Print two integers separated by spaces. The first one is the minimum possible money needed to ensure the security of all the junctions. And the second one is the number of ways you can ensure the security modulo 1000000007 ($10^9 + 7$).

Examples

input

Copy

```
3
1 2 3
3
1 2
2 3
3 2
```

output

Copy

```
3 1
```

input

Copy

```
5
2 8 0 6 0
6
1 4
1 3
2 4
3 4
```

Codeforces Round 244 (Div. 2)

Finished

→ Virtual participation



Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Problem tags

dfs and similar

graphs

two pointers

*1700

No tag edit access

→ Contest materials

- Announcement (en) 
- Tutorial (en) 

4 5	
5 1	
output	Copy
8 2	

input	Copy
10 1 3 2 2 1 3 1 4 10 10 12 1 2 2 3 3 1 3 4 4 5 5 6 5 7 6 4 7 3 8 9 9 10 10 9	
output	Copy
15 6	

input	Copy
2 7 91 2 1 2 2 1	
output	Copy
7 1	

